

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit “ 3 : Session “ 12 : SLO “ 1 | Circular Queue “ Dequeue Operation

■ I. Algorithm to Delete an Element from a Circular Queue

- Step 1: IF FRONT = -1
- Write UNDERFLOW
- Goto Step 4
- [END OF IF]
- Step 2: SET VAL = QUEUE[FRONT]
- Step 3: IF FRONT = REAR
- SET FRONT = REAR = -1
- ELSE
- IF FRONT = MAX-1
- SET FRONT = 0
- ELSE
- SET FRONT = FRONT + 1
- [END OF IF]
- [END OF IF]
- Step 4: EXIT

■ II. Programming Code “ delete_element()

```
• int delete_element() {  
• int val;  
• if (front == -1) {  
• printf("\n UNDERFLOW");  
• return -1;  
• }  
• val = queue[front];  
• if (front == rear)  
• front = rear = -1;  
• else {  
• if (front == MAX-1)  
• front = 0;  
• else  
• front++;  
• }  
• return val;  
• }
```